

# NOTES OF Star Formation Physics

version 1.0

Giacomo Pannocchia

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*For those who come after*



## PREFACE

*In a sense, you might say I'm getting quite fond of writing these kind of notes.*

Not long ago, it occurred to me how cool it is when someone unexpectedly releases a very detailed and all-comprehensive version of their notes, especially when dealing with a course that has a handful of topics often interacting together in unpredictable, yet fascinating, ways.

These notes will be mainly based on *my* own notes of the lectures by Professor F. Fontani and M. Padovani.

These notes refer to the lectures of the academic year 2026-2027, so future versions of these lectures might differ slightly. However, since I take little to no pride in my messy notes, I'll be often using some of the references you can find on the course catalogue page or that have been cited or shown during class. Either way, I'll do my best keeping track of them all in the bibliography of this humble collection.

You can report errors (whatever their nature might be) and suggestions for additions at [g.pannocchia3@studenti.unipi.it](mailto:g.pannocchia3@studenti.unipi.it) or through whatever convoluted way (conventional or not) you prefer<sup>1</sup>.

You can find all the notes I've redacted [here](#).

Without further ado, we'd better not lose much more time on a preface and get started with it.

*There was Eru, the One, who in Arda is called Ilúvatar; and he made first the Ainur [...] But for a long while they sang only each alone, or but few together, while the rest hearkened; for each comprehended only that part of the mind of Ilúvatar from which he came, and in the understanding of their brethren they grew but slowly. Yet ever as they listened they came to deeper understanding, and increased in unison and harmony.*

*Ainulindalë, "The music of the Ainur",  
Silmarillion, J. R. R. Tolkien*

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<sup>1</sup> I'd like, however, not to see my house stormed by homing pigeons.



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For this part of this chapter, I'll be mainly following [4], Chapters 1-2, but I won't fail to report other used references were I to do so.

The interstellar medium, or ISM, is, arguably, also the most important component of galaxies, for it is the ISM that is responsible for forming the stars that are the dominant sources of energy. While it now appears that the mass of most galaxies is primarily in the form of dark matter particles that are collisionless, or nearly so, it is the baryons (accounting for perhaps  $\sim 10\%$  of the total mass) that determine the visible appearance of galaxies, and that are responsible for nearly all of the energy emitted by galaxies, derived from nuclear fusion in stars and the release of gravitational energy in accretion disks around black holes.

The ISM has mainly four constituents:

1. Gas, which covers a whopping 99% of the total mass of the ISM. Note that when we refer to gas, we're imagining it as a macroscopic fluid/-body;
2. Dust, which has a significantly lower density than gas  $\rho_d \sim 1\%\rho_g$ . Dust is made of a hefty amount of *refractory elements*, mainly coming in the form of grains of different morphology and diameters in the range  $D \sim 0.1 - 10 \mu\text{m}$ . Dust is mostly present in cold environments, but incidentally also in regions where  $T \sim 10^5 - 10^6$ ;
3. Cosmic Rays are charged, highly energetic particles produced in extreme environments, like supernovae and the stellar winds of O/B-class stars. The vast majority of CR are protons ( $\sim 90\%$ ), followed by  $\alpha$  particles and electrons. Both have velocities much greater than those typical of the ISM: The most energetic ever detected had an energy of  $\sim 10^{21}$  eV. As such, they can easily penetrate even the most dense regions of the ISM.
4. Photons from many sources, including the cosmic microwave background (CMB) in the millimeter/radio band, stellar photospheres (i.e., starlight) in the optical band, radiation emitted by interstellar ions, atoms, and molecules in the infrared band; thermal emission from interstellar grains that have been heated by starlight still in the infrared. Then moving to the higher energy ( $\gamma$  and X) we also observe photons from free-free emission ("Bremsstrahlung") from interstellar plasma, synchrotron radiation from relativistic electrons, and gamma rays emitted in nuclear transitions and  $\pi_0$  decays.

*With the term refractory elements we're referring to solid particles that evaporate/sublimate at temperatures  $T \gg 1000$  K. For the sake of simplicity, they're usually assumed to be spherical.*

The mass of the gas component alone makes about 10% of the total mass of a galaxy, but it still is lower than the mass "stored" through stars. That said,

the ISM is far more spatially extended than stars ( $\sim 25 - 27$  kpc). Typical conversion rates  $\text{gas} \longleftrightarrow \text{stars}$  are

$$\frac{dM}{dt}(\text{gas} \rightarrow \text{stars}) \sim 1 - 5 M_{\odot} \text{ yr}^{-1} \quad \frac{dM}{dt}(\text{stars} \rightarrow \text{gas}) \sim 1 M_{\odot} \text{ yr}^{-1}$$

The baryons in the interstellar medium of the Milky Way are found with a wide range of temperatures and densities; because the interstellar medium is dynamic, all densities and temperatures within these ranges can be found somewhere in the Milky Way. However, it is observed that most of the baryons have temperatures falling close to various characteristic states, or “phases.” For purposes of discussion, it is convenient to name these phases. Here we identify seven distinct phases<sup>1</sup> that, between them, account for most of the mass and most of the volume of the interstellar medium. The seven phases are defined based on the temperature, density, and ionization status of Hydrogen atoms. These phases consist of the following:

1. *Hot ionized medium (HIM) or Coronal Gas:* Gas that has been shock-heated to temperatures  $T > 10^6$  K by blastwaves racing outward from supernova explosions. The gas is collisionally ionized, with ions such as OVI (i.e.  $\text{O}^{5+}$ ) present. Most of the coronal gas has low density ( $n < 0.01 \text{ cm}^{-3}$ ), filling an appreciable fraction (10 – 50%) of the volume of the galactic disk. The coronal gas regions may have characteristic dimensions of  $\sim 20$  pc, and may be connected to other coronal gas volumes. The coronal gas cools on  $\sim \text{Myr}$  time scales. Much of the volume above and below the disk is thought to be pervaded by coronal gas. The main heating mechanisms are (in individual galaxies) supernovae blast waves, (in galaxy clusters) release of gravitational energy, while they cool through X/ $\gamma$ -rays emission. The main tracers for this region are X/ $\gamma$ /UV-rays emission and radio emission from synchrotron radiation;
2. *Warm Ionized  $\text{H}_{\text{II}}$  Medium:* Gas where the hydrogen has been photoionized by ultraviolet photons from hot stars. Most of this photoionized gas is maintained by radiation from recently formed hot massive O-type stars - the photoionized gas may be dense material from a nearby cloud (in which case the ionized gas is called an  $\text{H}_{\text{II}}$  region) or lower density “intercloud” medium (referred to as diffuse  $\text{H}_{\text{II}}$ ). The density of the WIM depends on the particular environment (but roughly  $n \sim 1 \text{ cm}^{-3}$ ), but has a temperature of about  $T \sim 10^4 - 10^5$  K and fills about 10 – 15% of the volume of the galactic disk. The main heating mechanism is UV radiation from O-B stars, while they cool through high-excitation transitions of atoms. The main tracers for this region are radio lines and continuum (recombination lines, free-free), optical/UV lines;
3. *Warm neutral medium (WNM):* Predominantly atomic gas heated to temperatures  $T \approx 10^{3.7}$  K. In the local interstellar medium, this gas is found at densities  $n \approx 0.6 \text{ cm}^{-3}$ . It fills a significant fraction of the volume of the disk, perhaps 40%. The main heating mechanism is

*The Sun is located in the (warm) Local Bubble, a cavity with diameter of  $\sim 300$  ly, which could have been created by the explosion of a nearby Supernova (Geminga), 300,000 yr ago*

<sup>1</sup> Although how well-separated these phases are is a completely different question.

photoelectric effect on grains by photons, while cooling occurs mainly through the 21 cm line of atomic Hydrogen or through the  $158\ \mu\text{m}$  line of ionized Carbon, which are also the main tracers;

4. *Cold neutral medium (CNM)*: Predominantly atomic gas at temperatures  $T \approx 10^2\ \text{K}$ , with densities  $n \approx 30\ \text{cm}^{-3}$  filling  $\sim 1\%$  of the volume of the local interstellar medium. The main heating mechanisms are the interaction with cosmic rays and photoelectric effect on grains by photons, while cooling occurs mainly through the 21 cm line of atomic Hydrogen, which is also the main tracer;
5. *Molecular gas*: Two different phases can be distinguished here: the *diffuse molecular gas* and the *dense molecular gas*.
  - *Diffuse molecular gas*: similar to the cool  $\text{H I}$  clouds, but with sufficiently large (column) densities so that the self-shielding of  $\text{H}_2$  allows  $\text{H}_2$  molecules to be abundant in the cloud interior. It has a temperature of  $T \sim 50\ \text{K}$  and a density  $n \sim 100\ \text{cm}^{-3}$  and fills about 0.1% of the total galactic volume. The main heating mechanism is the interaction with cosmic rays, while cooling occurs mainly through the lines of atomic metals. The main tracer is molecular Hydrogen (but in absorption alone);
  - *Dense molecular gas*: Gravitationally bound clouds achieving  $n > 100\ \text{cm}^{-3}$ . These clouds are often “dark” - with visual extinction  $A_V \gtrsim 3\ \text{mag}$  through their central regions. In these dark clouds, the dust grains are often coated with “mantles” composed of  $\text{H}_2\text{O}$  and other molecular ices. It is within these regions that star formation takes place. They fill only the 0.001% of the total galactic volume, but make about 1/4 of the total mass. The main heating mechanism is the interaction with cosmic rays, while cooling occurs mainly through the roto-vibrational molecular lines of molecules, which are also the main tracers.

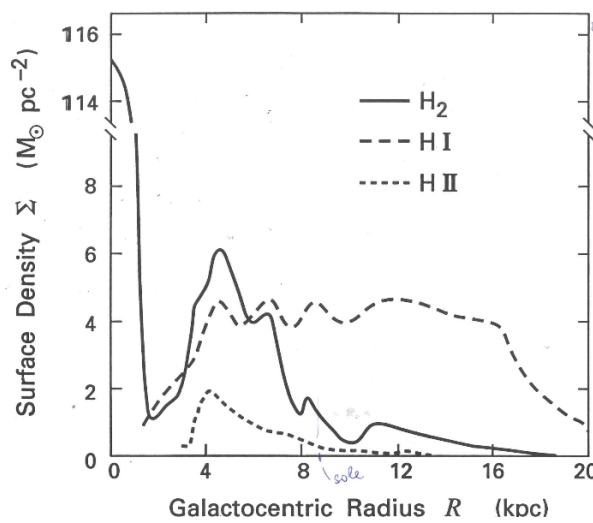


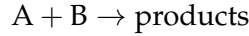
Figure 1.1: Distribution of different ionization states of Hydrogen as a function of the galactocentric radius. Credits: Unknown.

## 1.1 COLLISIONAL PROCESSES

Collisions are fundamental to the physics of the interstellar medium (ISM): They allow the gas to (usually) be treated as a fluid; they determine the thermal and electrical conductivity and diffusion coefficients; they produce most of the excitations of ions, atoms, and molecules that result in emission of photons from the ISM; and they are responsible for recombination of electrons and ions, and for chemical reactions.

### 1.1.1 Collisional Rate Coefficients

The rate per unit volume of a general two-body collisional process



is written as

$$r_{AB} = n_A n_B \langle \sigma v \rangle_{AB} \quad (1.1)$$

where the two-body collisional rate coefficient for the general collisional process is

$$\langle \sigma v \rangle_{AB} = \int_0^\infty \sigma_{AB}(v) v f_v dv \quad (1.2)$$

where  $\sigma_{AB}$  is the velocity velocity-dependent reaction cross section for the reaction and we're allowing the individual particles to have a non-trivial velocity distribution  $f_v dv$ . The collision rate coefficient thus includes two contributions: An "internal" part that takes into account both the physico-chemical properties of the colliders, and the quantum probabilities that a given interaction will produce a given product, and an "external" part that takes into account the kinetical status (i.e.  $v$ ) of colliders.

If we assume thermal equilibrium at temperature  $T$ , the distribution function for the relative speed in encounters between particles A and B is given by a Maxwellian velocity distribution

$$f_v dv = 4\pi \left( \frac{\mu}{2\pi kT} \right)^{3/2} \exp \left( -\frac{\mu v^2}{2kT} \right) v^2 dv \quad (1.3)$$

where  $\mu$  is the reduced mass of the collision partners. Note that if the velocity distribution of the individual collision partners is a Maxwellian distribution, the (relative) velocity distribution is still a Maxwellian and has the form (1.3).

Let us give a brief overview of the collisional rate coefficient for different types of interaction.

#### *Interaction between neutral species*

The main forces at play between atoms, grains, and neutral molecules are long-range forces (attractive forces between instantaneous dipoles) and close-range forces (repulsive Coulombian forces of electron). This configuration is easily approximated by the interaction between rigid spheres, allowing us to find an expression with the likes of

$$\langle \sigma v \rangle_{nn} = 1.81 \cdot 10^{-10} \left( \frac{T}{100 \text{ K}} \right)^{1/2} \left( \frac{m_H}{\mu} \right)^{1/2} \left( \frac{r_A + r_B}{2 \text{ \AA}} \right)^2 \text{ cm}^{-3} \text{ s}^{-1} \quad (1.4)$$

### Interaction between neutral and charged species

This is the case of an atom or neutral molecule interacting with an ion or electron. The relevant forces at play here are the attractive dipolar forces, which are non-negligible even at long distances. Unfortunately, this means that the rigid sphere approximation is no longer valid.

We should then consider the force exerted by the ion on the induced dipole

$$F = -\frac{2\alpha Z^2 e^2}{r^5} \quad (1.5)$$

to which is associated an interaction potential

$$U(r) = -\frac{1}{2} \frac{\alpha Z^2 e^2}{r^4} \quad (1.6)$$

A proper discussion should then consider the impact parameter for the process and deduce from there the collisional rate coefficient, which we'll here present without proof

$$\langle \sigma v \rangle_{nc} = 8.98 \cdot 10^{-10} Z \left( \frac{\alpha}{a_0^3} \right)^{1/2} \left( \frac{m_H}{\mu} \right)^{1/2} \text{ cm}^{-3} \text{ s}^{-1} \quad (1.7)$$

Notably, it is independent of the temperature! So, in a cold gas, neutral-charged interactions have a higher collisional rate coefficient than the neutral-neutral scenario.

## 1.2 STATISTICAL MECHANICS

The interstellar medium (ISM) and intergalactic medium (IGM) are generally far from thermodynamic equilibrium. Nevertheless, the methods of statistical mechanics and thermodynamics provide powerful tools for understanding the non-equilibrium conditions that prevail, and for relating forward and reverse rates for the processes, such as ionization and recombination, that shape the medium.

But before getting any deeper than this, we should ask whether it is sensible of us to assume the ISM like a fluid. The general definition of fluid—which doesn't trouble the poor Knudsen and his number—is that of a macroscopic body with a bulk velocity. To achieve such a specific condition, the system we're trying to probe must have a negligible stochastic dispersion with respect to the bulk velocity on a certain scale. We'll look into those scales that satisfy  $\lambda \gg l = 1/n\sigma$ , where  $l$  is the mean free path of the particles gathered in the wannabe fluid. As a rule of thumb, you might want to use the following expression

$$l = \frac{1}{\sigma} = 55 \left( \frac{r}{1 \text{ Å}} \right)^{-2} \left( \frac{n}{1 \text{ cm}^{-3}} \right)^{-1} \text{ AU} \quad (1.8)$$

### 1.2.1 Partition Function and Levels Population

Consider some physical system (e.g., atoms in a box of volume  $V$ ) that is able to exchange energy with a "heat reservoir" at temperature  $T$ . The theory of statistical mechanics defines the partition function  $\mathcal{Z}$  to be

*Note that we're implicitly assuming the particles to be identical and the system to be isolated.*

$$\mathcal{Z} = \sum_i g_i e^{-E_i/kT} \quad (1.9)$$

We know that the total number of particles  $N$  (which is fixed) must be equal to the sum of all the particles in the different energy states

$$N = \sum_i N_i \quad (1.10)$$

It is fairly easy, although tedious, to show that the number of particles in the  $i$ -th energy state is given by

$$N_i = \frac{N}{\mathcal{Z}(T)} g_i e^{-E_i/kT} \quad (1.11)$$

This configuration is crucial for the establishing of Local Thermodynamic Equilibrium: Matter obeys to the Boltzmann distribution for energy and Maxwell distribution for velocity, but LTE means that matter is not in *global* equilibrium with radiation interacting with it.

What if now we relax the condition of the particles being identical? The process grows immediately more tedious and verbose, but the procedure to follow is essentially the same.

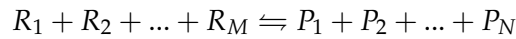
If we get an isolated system with total energy  $E$ , total mass  $\mathcal{M}$ , with  $M$  reactant species,  $N$  product species: the most likely macroscopic state is that with maximum number of compatible microscopic states, that is the one that maximizes entropy. This time we need to find the ways in which the particles of each species can be combined.

Skipping all the calculations we find, instead of the number of particles, the volume densities that maximize the entropy

$$\frac{\prod_{i=1}^N n_{P_i}^{b_i}}{\prod_{j=1}^M n_{R_j}^{b_j}} = \left[ \frac{(2\pi kT)^{3/2}}{h^3} \right]^{\sum a_i - \sum b_j} \left[ \frac{\prod_{i=1}^N M_{P_i}^{b_i}}{\prod_{j=1}^M M_{R_j}^{b_j}} \right]^{3/2} \frac{\prod_{i=1}^N z_{\text{int}}(P_i, T)^{b_i}}{\prod_{j=1}^M z_{\text{int}}(R_j, T)^{a_j}} \quad (1.12)$$

which is the *law of mass action*. At equilibrium, the number of particles per unit volume ( $n = N/V$ ) of each species involved is related only to temperature and to the partition functions of the individual species. The law is valid for every reaction or collision that happens in LTE.

Assume a reaction



so that the ratio between the collision rate coefficients is

$$\frac{\langle \sigma v \rangle_{\leftarrow}}{\langle \sigma v \rangle_{\rightarrow}} = \frac{\prod_{i=1}^M n_{R_i}}{\prod_{j=1}^N n_{P_j}}$$

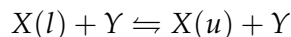
so that plugging in the law of mass action

$$\left( \frac{\langle \sigma v \rangle_{\leftarrow}}{\langle \sigma v \rangle_{\rightarrow}} \right)^{-1} = \left[ \frac{(2\pi kT)^{3/2}}{h^3} \right]^{N-M} \left[ \frac{\prod_{i=1}^N M_{P_i}^{b_i}}{\prod_{j=1}^M M_{R_j}^{b_j}} \right]^{3/2} \frac{\prod_{i=1}^N z_{\text{int}}(P_i, T)^{b_i}}{\prod_{j=1}^M z_{\text{int}}(R_j, T)^{a_j}} \quad (1.13)$$

Perhaps I'm tripping (fairly possible), but the law of detailed balance seems to be just the reciprocal of (1.12).

results in the law of detailed balance.

This is fairly useful when applied to a realistic scenario, such as that of collisional (de)excitation

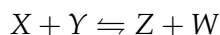


which has

$$\left( \frac{\langle \sigma v \rangle_{ul}}{\langle \sigma v \rangle_{lu}} \right) = \frac{z_{\text{int}}(X(l), T)}{z_{\text{int}}(X(u), T)} \implies \left( \frac{\langle \sigma v \rangle_{ul}}{\langle \sigma v \rangle_{lu}} \right) = \frac{g_l}{g_u} \exp(E_{ul}/kT) \quad (1.14)$$

The rate coefficients for a process of (de-)excitation of a given species are regulated by: (i) Temperature, (ii) degeneracy of involved levels, (iii) their energy difference.

Another example might be that of a generic chemical reaction



which has

$$\frac{\langle \sigma v \rangle_{\leftarrow}}{\langle \sigma v \rangle_{\rightarrow}} = \frac{z_{\text{int}}(X, T) z_{\text{int}}(Y, T)}{z_{\text{int}}(Z, T) z_{\text{int}}(W, T)} \propto \frac{\exp(-(E_X + E_Y)/kT)}{\exp(-(E_Z + E_W)/kT)} \quad (1.15)$$

### 1.3 ENERGY LEVELS OF MOLECULES

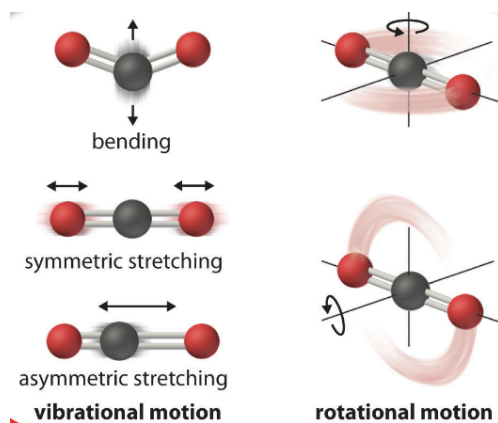
To properly discuss the formation and "position" of the energy levels of a given molecule, we should consider, in principle, quite a lot of contributions: The kinetic term, the attractive potentials of each atom, the electronic repulsion, and the electrostatic repulsion between the atoms themselves. The resulting Hamiltonian is rather complicated to treat as it is.

We then introduce the *Born-Oppenheimer approximation*, which assumes that Nuclei are essentially fixed and the electrons are free to move: The result is that the motion of the nuclei does not induce electronic transitions due to the different mass of electrons and nuclei. Under this approximation, you can demonstrate

$$E_{\text{tot}} = E_{\text{el}} + E_{\text{vib}} + E_{\text{rot}}$$

As a further approximation, we'll assume that centrifugal distortion is negligible and only symmetric stretching is allowed.

$E_{\text{el}}$  is the energy of electronic state,  $E_{\text{vib}}$  energy due to radial vibration of nuclei with respect to the equilibrium position,  $E_{\text{rot}}$  energy of rotation of nuclei about the centre of mass.



Like in atoms, also in molecules the electronic transitions emit in the optical/UV bands, while the roto-vibrational transitions fall in the radio/m-m/infrared bands. Note, however, that at the temperature of the molecular gas ( $T < 100\text{ K } 10^{-2}\text{ eV}$ ) only the rotational levels in the ground vibrational and electronic states are populated!

The relevant parameters we'll need in our discussion are

- The classical energy  $E$  of a rigid rotator

$$E = \frac{J_a^2}{2I_a} + \frac{J_b^2}{2I_b} + \frac{J_c^2}{2I_c} \quad J^2 = J_a^2 + J_b^2 + J_c^2$$

where  $a$ ,  $b$  and  $c$  are the principal axis of the rotator;

- The permanent dipole moment  $\mu$ . If the dipole moment is non-zero, electric dipole transitions are allowed. On the other hand, if the dipole moment is vanishing, only electric quadrupole transitions (and on-wards) are allowed. These, however, are strongly disfavored.

In the following, we'll consider these configurations

- Diatomic, linear:  $I_a = I_b \neq I_c = 0$ ;
- Symmetric-top:  $I_a = I_b \neq I_c \neq 0$ ;
- Asymmetric-top:  $I_a \neq I_b \neq I_c$ ;
- Spherical:  $I_a = I_b = I_c$

### Linear Rotors

The classical energy is easily expressed in terms of the total (square) angular momentum  $E = J^2/2I$ , which provides us a simple expression for the quantum version. Hence

$$E_j = B_{\text{rot}} h j(j+1) \quad B_{\text{rot}} = \frac{h}{8\pi^2 I}$$

and selection rules force transitions with  $\Delta j = \pm 1$ . Note that from (1.3) we deduce that  $E_j \propto j^2$  and  $h\nu = E_{j+1} - E_j \propto j$ . Also note that  $E_j \propto B_{\text{rot}} \propto I^{-1}$ , hence species with larger  $B_{\text{rot}}$  (at fixed  $j$ ) have larger  $E_{j+1}$ ,  $E_j$  and  $\nu$ .

This has a rather troublesome implication: Molecular Hydrogen is only seen through absorption lines, and rarely from emission lines. This is for two reasons:

1. Molecular Hydrogen is *omonuclear*, so no dipole transitions but only electric quadrupole transitions  $\Delta j = \pm 2$ ;
2. Molecular Hydrogen has *minimum moment of inertia*, so a huge separation between levels, and high  $T$  is needed to populate the excited states. Roto-vibrational transitions are visible only in very hot gas.

The lowest energy quadrupole transition for molecular Hydrogen has  $\Delta E = 510\text{ K}$ .

### Symmetric-top Rotors

For the sake of simplicity, let us define  $I_a = I_b = I_{\perp}$ , and  $I_c = I_{\parallel}$ . The classical energy is thus written as

$$E = \frac{J^2}{2I_{\perp}} + J_c^2 \left( \frac{1}{2I_{\parallel}} - \frac{1}{2I_{\perp}} \right)$$

To the two quantum operators  $J^2$  and  $J_c$  are associated the eigenvalues  $j(j+1)\hbar^2$  and  $m\hbar$ , with  $m \leq |j|$ . We can define two rotational constants

$$A = \frac{h}{8\pi^2 I_{\parallel}} \quad B = \frac{h}{8\pi^2 I_{\perp}}$$

which we can use to write the rotational energy

$$E_{j,m} = Bhj(j+1) + (A-B)hm^2 \quad h\nu = 2hBj \quad (1.16)$$

and selection rules force transitions with  $\Delta j = \pm 1$  and  $\Delta m = 0$ .

### Asymmetric-top Rotors

This case is somewhat complicated and simple at the same time, because a general analytical expression that holds for all asymmetric-top rotators for  $E_j$  and  $\nu_{ij}$  does not exist. The rotational energy will depend, not surprisingly, on two more quantum numbers, and we have to define three rotational constants.

#### Example: H<sub>2</sub>O

The molecule of water has the two atoms of Hydrogen form an angle of  $104.45^\circ$ . It has a blatant axis of symmetry passing from the atom of Oxygen, thus meaning the molecule has three different momenta of inertia.

It is crucial to distinguish two possible configurations: *ortho* (the two Hydrogen atoms have parallel spins) and *para* water (the two Hydrogen atoms have anti-parallel spins). The complete wave function for the molecule of water (which is antisymmetric) is the product of four contributions  $\psi = \psi_{el}\psi_{vib}\psi_{rot}\psi_s$ , where the former two are symmetric, while the latter two must always have opposing parity.

- Ortho:  $m_{-1}, m_{+1} = \text{even, odd (eo) or odd, even;}$
- Para:  $m_{-1}, m_{+1} = \text{even, even (ee) or odd, odd;}$

but since selection rules force  $\Delta j = 0, \pm 1$  and  $\Delta m = \pm 1, \pm 3$ , only transitions  $eo \rightarrow oe$  or  $ee \rightarrow oo$  (and their inverse) are allowed, while transitions between ortho and para states are strongly forbidden.

### Spherical Rotors

The expression this time is fairly simple

$$E = \frac{J^2}{2I} \implies E_j = \frac{j(j+1)\hbar^2}{2I} \quad h\nu = \frac{(j+1)\hbar^2}{I} \quad (1.17)$$

Note that non-polar molecules have no rotational dipolar lines unless they're distorted. Distortion, however, is only relevant for high values of  $j$ .

### 1.3.1 Lines Hyperfine Structure

Some nuclei have non-zero nuclear spin, which interacts (either via magnetic or electric quadrupole momentum) with electrons and/or the other nuclear spins. The energetic splitting due to this interaction gives the *line hyperfine structure*. Only some nuclei have non-zero nuclear spin, indicated usually with quantum number  $I$ , which depends on the particular combination of number of protons and neutrons in the nucleus. For example, the molecule  $\text{C}^{17}\text{O}$  has the  $^{17}\text{O}$  nucleus with nuclear spin  $I = 5/2$  and presents the splitting shown in Fig.1.2.

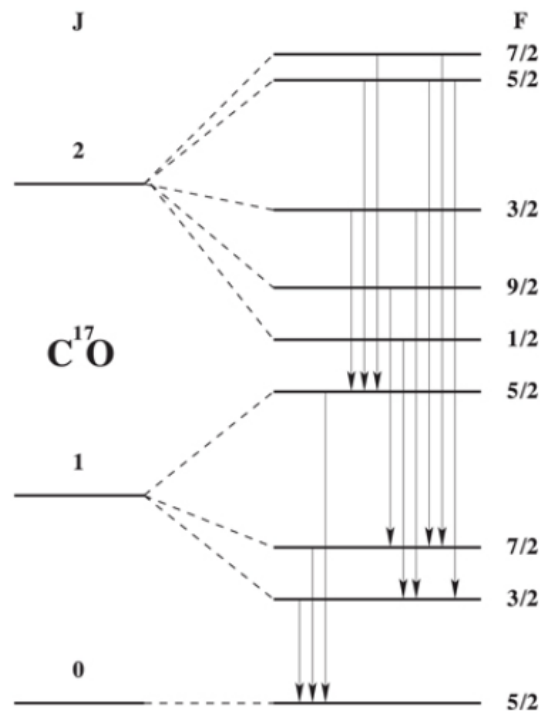


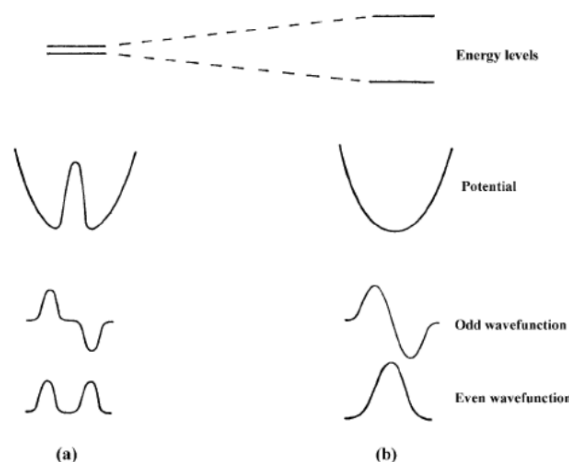
Figure 1.2: Hyperfine splitting induced by nuclear spin on the  $\text{C}^{17}\text{O}$  molecule.

### 1.3.2 Inversion Transitions of $\text{NH}_3$

To address what's commonly known as inversion transition, as far as the molecule of Ammonia is concerned, we need mainly three elements:

1. *Motion*: Vibration of the N nucleus that goes up and down the plane of H nuclei;
2. *Associated potential*: The harmonic potential with a barrier centred on  $z = 0$ . The N nucleus can penetrate the barrier via quantum tunnelling;
3. *States*: The two quantum states at lowest energy associated with this potential, indicated as + and -, are the equivalent of those of the barrierless harmonic potential, and each rotational state defined by quantum

numbers (J, K) ( $\text{NH}_3$  is symmetric-top) is splitted in the two inversion states + and -



**Figure 1.3:** Development of energy level splitting when the potential barrier is reduced. (a) With barrier present, the wavefunctions have similar forms and energy levels are closely spaced; (b) as barrier is reduced, wavefunctions take on distinctly different form and energy levels separate.

The two inversion levels + and - are further split into hyperfine levels due to the non-zero spin of the N nucleus and of the three H nuclei. So, for example, in the rotational state  $(j, k) = (1, 1)$  the two inversion levels split into 5 hyperfine sub-levels; in total, the inversion transition (1,1) is splitted into 18 hyperfine components.

### Vibrational Levels

In principle, we should also consider the energy level produced by vibration motions of biatomic molecules. Classically, we know that at large separations the Coulombian attraction of electrons on nuclei is dominating, while, on the shorter scales, prevails the Coulombian repulsion of nuclei. For biatomic molecules, the best approximation for this particular potential is the *Morse potential*

$$V(r) = D_e(1 - \exp(-a(r - r_e)))^2 \quad (1.18)$$

which we can Taylor-expand in a neighborhood of  $r_e$ , finding that the first non-vanishing order is the second one

$$V(r) = \frac{1}{2}k(r - r_e)^2 + o(r^3) \quad k = \left( \frac{\partial^2 V}{\partial r^2} \right)_{r=r_e} = 2a^2 D_e \quad (1.19)$$

and so the Morse potential is equivalent to an harmonic potential, which we know fairly well how to treat, also quantistically speaking. Note that unharmonic contributions grow (significantly) relevant moving to larger distances from the equilibrium position.



## 2 | INTRODUCTION TO ASTROCHEMISTRY

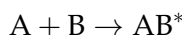
Thanks to the advent and refining of radiotelescopes throughout the past century, it was possible to identify new molecules, both organic and inorganic, all around the ISM of the MW and, sporadically, in external galaxies (which is a considerable boon).

As of September 2025,  $\sim 330$  molecules were identified in the Milky Way and  $\sim 74$  in external galaxies. You can find an extensive list, if you're interested, at this [link](#).

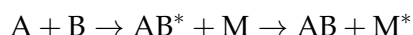
Not surprisingly, we've been detecting molecules of the most abundant and (chemically) active elements, that are Oxygen, Carbon, and Nitrogen.

### 2.1 SURFACE CHEMISTRY

The most elemental process is the radiative association of two atoms, A e B to form the molecule AB. The molecule will have an internal energy



The two associating atoms form a compound known as activated complex ( $AB^*$ ) which will usually radiate away the excess energy produced by its formation.  $AB^*$  must lose this excess energy during the impact to remain bound and not dissociate immediately. This energy loss through emission of photons is a *very rare process*. However, the molecule can lose this excess energy via three-body collisions



which are still rather rare in the ISM, where density is at most  $\sim 10^7 - 10^{10} \text{ cm}^{-3}$ . De-excitation through three-body collisions is much more probable in denser environments like Earth's atmosphere.

So how can we prevent the radiative dissociation of molecules in the ISM? Dust grains, thanks to their very large surface and heat capacity, can favor atom encounters and easily absorb the excess energy.

The process goes through three fundamental steps, which are also pictorially shown in Fig.2.1:

1. *Accretion*: the particle impacts the grain surface and can be *adsorbed*<sup>1</sup> if the temperature is lower than the evaporation temperature. Accretion can occur via two dominant mechanisms
  - *Physisorption*: van der Waals forces (0.01 – 0.2 eV);
  - *Chemisorption*: chemical bond (1 eV).

<sup>1</sup> i.e. can stick on surface. And no, it's not a typo.

2. *Surface diffusion*: if the particle is not chemisorbed, it can move on the grain surface, meet other species and, maybe, react to form new molecules.
3. *Desorption*: if temperature increases above the evaporation temperature, the new molecule goes in the gas.

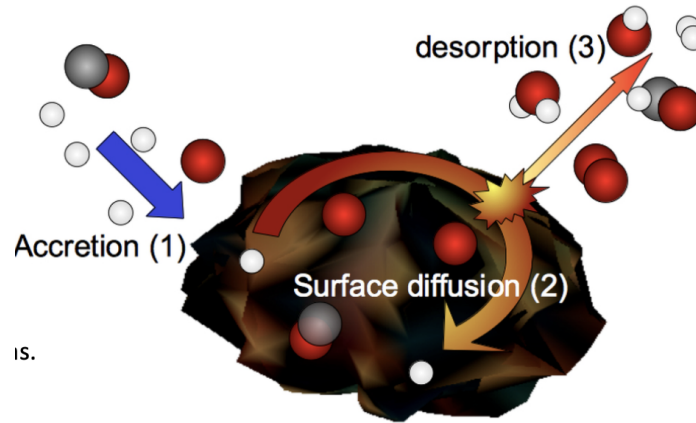
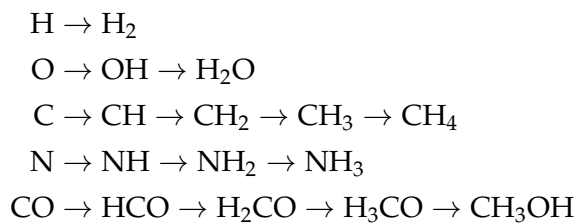


Figure 2.1: The three fundamental steps of surface chemistry on dust grains.

The “sites” where the probability to form new molecules is highest are the gravitational potential walls on the (non-smooth) dust grains. Transition from one site to the other can happen via two mechanisms: Thermal hopping<sup>2</sup> (the adsorbed species migrate from one site to the other exploiting the warming of the dust grain), and the good old quantum tunneling.

Hydrogen (and Deuterium) is the most abundant and most “mobile” reactive species. Therefore, hydrogenation (and deuteration) is a very efficient surface chemistry process



If you’re interested in the subject, you might want to take a look at [1], [3], [7], [9], and many others for which I didn’t manage to pinpoint the exact article.

## 2.2 GAS-PHASE CHEMISTRY

After  $\text{H}_2$  was formed and evaporated from grains, the gas-phase chemistry starts.  $\text{H}_2$  is so much more abundant than any other molecule that if a species can react with  $\text{H}_2$ , no other reactions need to be considered. How

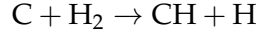
<sup>2</sup> You can read, for example, [6].

can we understand if a reaction with  $\text{H}_2$  is spontaneous? A reaction at  $p$  and  $T$  constant is spontaneous if the Gibbs' free energy

$$\Delta G = \Delta H - T\Delta S = \Delta U + RT\Delta N - T\Delta S < 0 \quad (2.1)$$

In particular, a reaction is spontaneous at low temperature if  $\Delta U < 0$ , hence only molecular binding energy is involved.

Consider for example the reaction



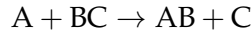
Molecular Hydrogen has a binding energy of 4.48 eV, while CH has 3.47 eV. Hence, the reaction is endothermic  $\Delta U = U_r - U_p > 0$  and cannot occur spontaneously in molecular clouds, where typically  $kT < 0.01$  eV.

In the molecular ISM, only exothermic reactions can occur!

*Note that this condition holds until the gas is isolated. Clearly, energy injection from external factors might provide the energy needed by endothermic reactions to start.*

### 2.2.1 Neutral-Neutral Reactions

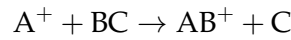
Consider a reaction like



Reactions of this sort present an *activation energy*, which is the minimum energy to break the existing bond. Usually, this energy is of the order of  $E_a \sim 0.11 - 1$  eV while the temperature of the ISM is  $T \lesssim 0.01$  eV. This means that only exothermic and barrierless reactions can occur in the molecular ISM.

### 2.2.2 Neutral-Charged Reactions

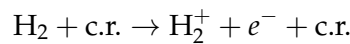
Also called Ion-Neutral, they involve interactions of the sort of



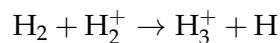
Ion-neutral exothermic reactions typically do not have activation barrier, thanks to the long range attractive force between the ion and the induced dipole moment (see Chapter 1, or [2] and [5]). This is because the collision rate coefficient has no dependence from the temperature of the region for this particular type of interaction.

## 2.3 FORMATION OF $\text{H}_3^+$

After the formation of molecular Hydrogen. cosmic rays can ionize the molecule

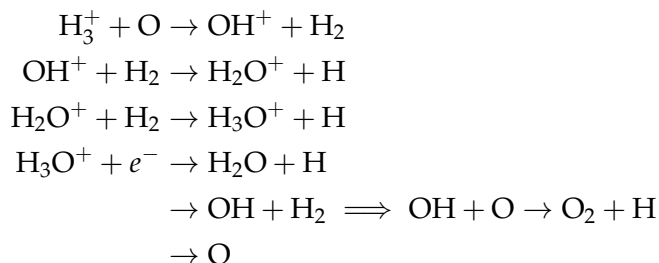


Once  $\text{H}_2^+$  is formed in small amounts, it reacts very quickly with the very abundant  $\text{H}_2$  to form  $\text{H}_3^+$  the most abundant and important molecular ion in the ISM



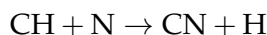
This new ion triggers ion-neutral reactions that lead to the formation of more complex molecules. It can easily donate a proton to some neutrals via exothermic reactions, e.g. with atomic O, and produce another ion. If this ion can react with H<sub>2</sub>, it gives rise to another ion, and so on. The chain of reactions with H<sub>2</sub> stops when the final ion cannot react with H<sub>2</sub> anymore for energetic reasons.

For example



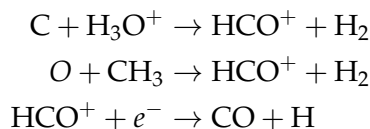
It can also prompt the formation of simple hydrocarbons: The formation of carbon-bearing species starts with a sequence of reactions very similar to those that initiate the chemistry of oxygen-bearing species. It proceeds through a series of *proton transfers* from H<sub>3</sub><sup>+</sup> to a neutral atom, forming a hydrogenated ion, then to *Hydrogen abstraction reactions*, which is a chain of reactions in which one H of H<sub>2</sub> is transferred to an ion. It stops to an ion that cannot react with H<sub>2</sub> anymore, and finally ends with *dissociative recombination*, the ion that cannot react with H<sub>2</sub> dissociates upon recombination with an electron.

Then again, H<sub>3</sub><sup>+</sup> also prompts the initiation of Nitrogen chemistry which, however, starts from

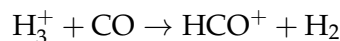


## 2.4 FORMATION AND DESTRUCTION OF CO

Carbon Monoxide is formed thanks to a series of chemical reactions

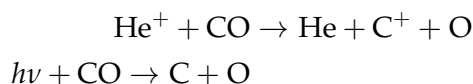


Carbon Monoxide is very stable ( $B \sim 11.09$  eV) and difficult to destroy. Curiously enough, it is destroyed by the very ion that prompts the start of all the fancy chemistry we've described in the last section

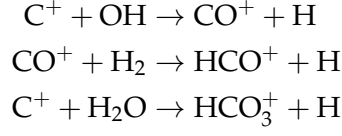


but the last reaction of the previous cycle immediately reforms CO.

More efficient ways to "remove" Carbon Monoxide are



but part of  $C^+$  reacts with OH and water (but not with molecular Hydrogen, since the reaction is endothermic)



while most photons able to dissociate CO (so with  $h\nu > 11.09 \text{ eV}$ ) are absorbed in advance from  $H_2$ .

## 2.5 FREEZE-OUT IN COLLAPSING CLOUDS

Let us consider a collapsing cloud: as long as the depletion timescale is longer than the free-fall timescale, the *freeze-out*<sup>3</sup> does not occur during collapse. But if  $\tau_{\text{dep}}$  is shorter than  $\tau_{\text{ff}}$ , the molecule freezes during collapse, disappearing from the gas! In the solid phase, it cannot be “seen” in emission anymore. In short, we have to compare

$$\tau_{\text{ff}} = \left( \frac{3\pi}{32G\rho} \right)^{-1/2} = 4 \cdot 10^{-7} n_{\text{H}}^{-1/2} \text{ yr} \quad (2.2)$$

with

$$\tau_{\text{dep}} = \frac{1}{\alpha n_d \sigma_d \bar{v}_t} \approx 10^9 \sqrt{\frac{m_X}{T}} (n_{\text{H}} \alpha)^{-1} \text{ yr} \quad (2.3)$$

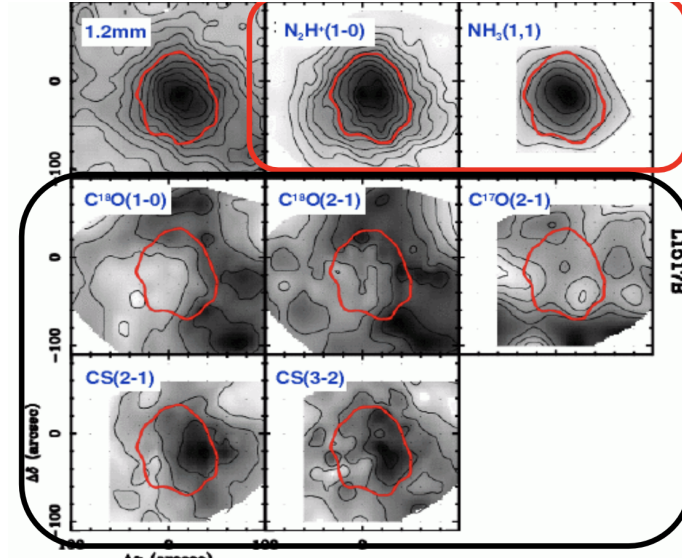


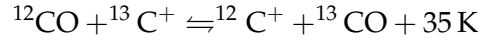
Figure 2.2: How a collapsing core looks like in the emission of gas and dust.

<sup>3</sup> Please, do not mistake this freeze-out with other kinds of freeze-outs, like neutrons freeze-out and the sorts. Here the freezing is quite literal, while in the other case is more of a figure of speech to hint at the fact that their abundance is no longer changing.

## 2.6 ISOTOPIC FRACTIONATION

We define *isotopic fractionation* the the chemical process(es) by which the fraction of heavier isotopes in molecules varies through chemical reactions.

For example

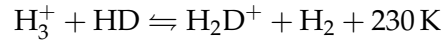


The difference in molecular binding energy between reactants and products, caused by different zero-point vibration energy, makes one direction exothermic, i.e. favoured, in cold gas: The molecule with heavier isotope has lower *zero-point energy*.

The zero-point energy of a biatomic molecule is usually expressed through the equation

$$E_v \approx \frac{1}{2}g \left( \frac{1}{2\pi} \sqrt{\frac{k}{m_r}} \right) \quad (2.4)$$

If temperature is lower than  $T \leq 20 \text{ K}$ , then



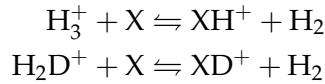
and the ratio  $\text{H}_2\text{D}/\text{H}_3^+$  increases of orders of magnitude since

$$\frac{n(\text{H}_3^+)}{n(\text{H}_2\text{D}^+)} \frac{n(\text{HD})}{n(\text{H}_2)} = \frac{\langle \sigma v \rangle_{\leftarrow}}{\langle \sigma v \rangle_{\rightarrow}}$$

which implies that

$$\frac{n(\text{H}_3^+)}{n(\text{H}_2\text{D}^+)} = \frac{n(\text{H}_2)}{n(\text{HD})} \exp \left( -\frac{\Delta E}{kT} \right)$$

On the other hand, if CO depletes ( $n \gtrsim 10^5 \text{ cm}^{-3}$ ),  $\text{H}_3^+$  and  $\text{H}_2\text{D}^+$  remain abundant



Both the conditions we presented are typical in dense and cold cores: The ratio  $\text{XD}^+/\text{XH}^+$  is orders of magnitude larger than the  $[\text{D}/\text{H}]$  ISM value ( $10^{-5}$ ).

Part I

APPENDIX



# 3

## APPENDIX

### APPENDIX B: BLACKBODY FUNCTION

Even at an undergraduate level, we're all fairly familiar with *blackbody radiation*. The easiest way to deduce the expression for the energy density of photons in *thermal equilibrium* (STE) inside a cavity is by the means of statistical mechanics.

Remember the Bose-Einstein distribution ( $\mu = 0$ )

$$n = \frac{1}{\exp(h\nu/kT) - 1}$$

and the phase space density of states (per unit volume)

$$\rho(\nu) d\nu = \frac{4\pi h g \nu^3}{c^3} d\nu$$

from which deducing the expression from internal energy is straightforward. Remembering  $g = 2$  is the quantum degeneracy of photons, a simple multiplication of the previous expressions yields

$$U_\nu d\nu = \frac{8\pi h}{c^3} \frac{\nu^3}{\exp(h\nu/kT) - 1} d\nu \quad (3.1)$$

The energy of a photon is  $E_\gamma = h\nu$ , meaning the number of photons in the frequency range  $\nu \rightarrow \nu + d\nu$  is

$$n_\nu d\nu = \frac{8\pi}{c^3} \frac{\nu^2}{\exp(h\nu/kT) - 1} d\nu \quad (3.2)$$

which can be integrated over all frequencies, thus obtaining the number density of photons in blackbody radiation

$$n_\gamma = \beta T^3 \quad \beta = \frac{2.4041k^3}{\pi^2 \hbar^3 c^3} \quad (3.3)$$

Since blackbody radiation is isotropic (it depends only on the absolute temperature  $T$ ), the definition of mean monochromatic intensity yields

$$B_\nu(T) = \frac{2h\nu^3}{c^2} \frac{1}{\exp(h\nu/kT) - 1} \quad (3.4)$$

It's important to notice that, in principle, such a fundamental result holds only in *strict thermodynamic equilibrium* (STE), but it's possible to generalize this formulation for less "restrictive" environments.

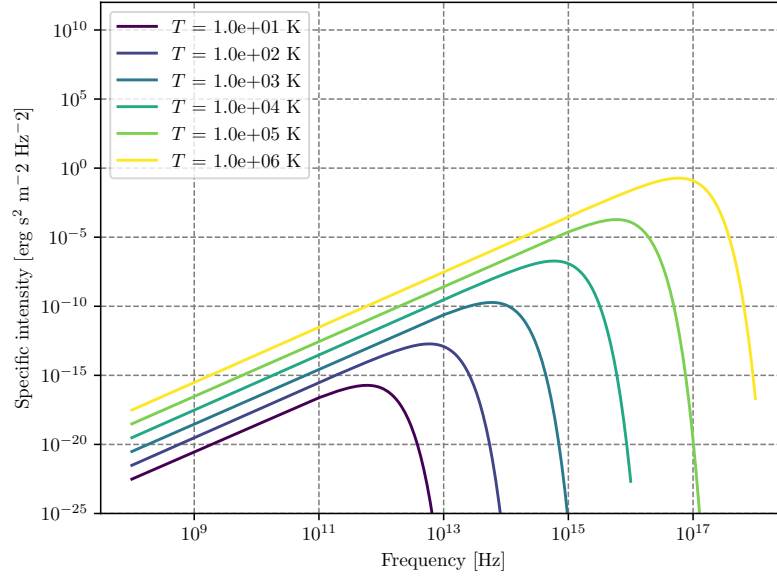


Figure 3.1: Blackbody frequency spectrum.

An incredible number of important results descends from (3.4), and it may be worthwhile to cite at least some of them, starting from Stefan-Boltzmann law. We'll use the following result without proving it

$$\int_0^{+\infty} B_\nu(T) d\nu = \frac{2h}{c^2} \frac{\pi^4}{15} \left( \frac{kT}{h} \right)^4$$

Computing the bolometric flux and the bolometric energy density by integrating over all frequencies using what we've just written down, you find the following

$$U(T) = aT^4 \quad F(T) = \sigma_{SB}T^4$$

Clearly the two constants  $a$  and  $\sigma_{SB}$  cannot be independent, and are actually related by the integral we've previously calculated. Using for example<sup>1</sup>

$$F(T) = \pi \int_0^{+\infty} B_\nu(T) d\nu$$

you can easily find out that the *Stefan-Boltzmann constant* is equal to

$$\sigma_{SB} = \frac{2\pi^5 k^4}{15c^2 h^3}$$

and the relation with  $a$  is simply  $\sigma_{SB} = ac/4$ .

The equation

$$F(T) = \frac{2\pi^5 k^4}{15c^2 h^3} T^4 \tag{3.5}$$

is what is usually known as the *Stefan-Boltzmann law*.

<sup>1</sup> The emergent flux from an isotropically emitting surface (such as a blackbody) is  $\pi \cdot$  brightness which is none other than the specific intensity.

Let us now consider two different regimes for eq.3.4:  $h\nu/kT \ll 1$  and  $h\nu/kT \gg 1$ . The first yields what is commonly known as the Rayleigh-Jeans Law which is, sadly, pretty much relevant only for radioastronomy.

Since

$$\exp\left(\frac{h\nu}{kT}\right) = 1 + \frac{h\nu}{kT} + o\left(\frac{h\nu}{kT}\right)^2$$

the blackbody radiation assumes the much simpler form of

$$B_\nu^{RJ} = \frac{2\nu^2}{c^2} kT$$

Another important results is achieved in the opposite regime, when the exponential term is rather larger than unity

$$B_\nu^W = \frac{2h\nu^3}{c^2} \exp\left(-\frac{h\nu}{kT}\right)$$

This expression is known as Wien's Law.

### Characteristic Temperatures

Starting from the blackbody spectrum we can give various definitions of temperature, that are going to be more and less useful when dealing with stars. The definitions are those of [8].

#### *Brightness Temperature*

One way of characterizing brightness (specific intensity) at a certain frequency is to give the temperature of the blackbody having the same brightness at that frequency. That is, for any value  $I_\nu$  we define  $T_b(\nu)$  by the relation

$$I_\nu = B_\nu(T_b)$$

this way of specifying brightness has the advantage of being closely connected with the physical properties of the emitter, and has the simple unit (K).

This procedure is used especially in radio astronomy, where the Rayleigh-Jeans law is usually applicable.

$$T_b = \frac{c^2}{2\nu k} I_\nu \quad h\nu/kT \ll 1 \quad (3.6)$$

Note that the uniqueness of the definition of brightness temperature relies on the monotonicity property of Planck's law. Also note that, in general, the brightness temperature is a function of  $\nu$ .

#### *Color Temperature*

Often a spectrum is measured to have a shape more or less of blackbody form, but not necessarily of the proper absolute value. For example, by measuring  $F_\nu$ , from an unresolved source we cannot find  $I_\nu$ , unless we know the distance to the source and its physical size.

By fitting the data to a blackbody curve without regard to vertical scale, a color temperature  $T_c$ , is obtained. Often the “fitting” procedure is nothing more than estimating the peak of the spectrum and applying Wien’s displacement law to find a temperature.

### *Effective Temperature*

The effective temperature of a source  $T_{eff}$  is derived from the total amount of flux, integrated over all frequencies, radiated at the source. We obtain  $T_{eff}$  by equating the actual flux  $F$  to the flux of a blackbody at temperature  $T_{eff}$

$$F = \int I_\nu \cos \theta \, d\nu \, d\Omega := \sigma T_{eff}^4 \quad (3.7)$$

## ABOUT ME

Hi there, I'm Giacomo, sometimes known as Cael by a very small niche of people, the author of the notes you just finished reading.

I'm a Bachelor-graduate student at the University of Pisa now in the process of undertaking a major (Master degree) in Astronomy and Astrophysics at the same university. In my (meager) free time I'm an avid reader, gamer, anime-enthusiast as well as an independent author in the very process of publishing his first novel.

For those of you that may be interested I'll leave both the link to the novel's page as well to my instagram profile in the QRcodes right below.

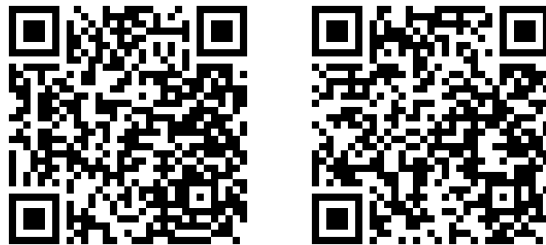


Figure 3.2: On the left is the link to my instagram profile, on the right the one to my novel's page (Python really has a library for anything, huh?).

Come say hi if you fancy action-fantasy stories with magic, swords, some comedy to garnish and a lot of other stuff :)

At last but not the least, I want to thank all my colleagues and fellow students that helped (and hopefully will help me again) during the drafting of this text, pointing out errors and occasions for further expansion.

If you'd like your names to be included and take your fair share of credits, feel free to contact me and I'll add your name right away in the few lines that are left of this page.

I hope my understading of the complex interplayings going on in our Universe has managed to facilitate your own understading.

If not, well... oopsie.

*See you, space cowboys...*

Giacomo



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